6.3.3 Problems involving Angle of Elevation and Depression

Let us consider the following situation.

A man standing at a top of lighthouse located in a beach watch on aeroplane flying above the sea. At the same instant he watch a ship sailing in the sea. The angle with which he watch the plane correspond to angle of elevation and the angle of watching the ship corresponding to angle of depression. This is one example were one oberseves both angle of elevation and angle of depression.

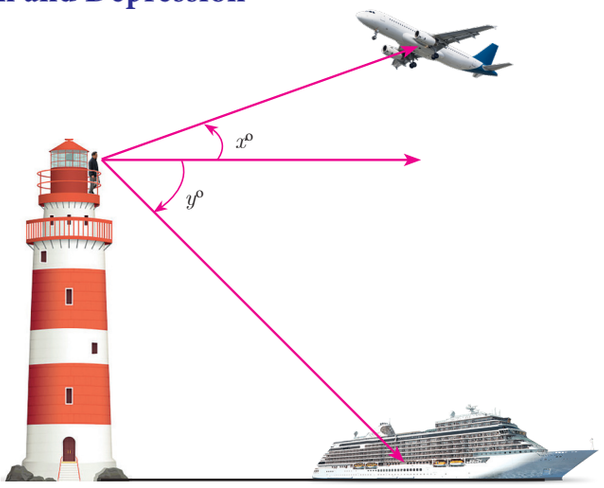


Fig. 6.26

In the Fig.6.26, $x°$ is the angle of elevation and $y°$ is the angle of depression.

In this section, we try to solve problems when Angles of elevation and depression are given.

Example 6.31 From the top of a 12 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30°. Determine the height of the tower.

Solution As shown in Fig.6.27, OA is the building, O is the point of observation on the top of the building OA. Then, $OA = 12$ m.

PP' is the cable tower with P as the top and P' as the bottom.

Then the angle of elevation of P, $\angle MOP = 60°$.

And the angle of depression of P', $\angle MOP' = 30°$.

Suppose, height of the cable tower $PP' = h$ metres.

Through O, draw $OM \perp PP'$

$MP = PP' - MP' = h - OA = h - 12$

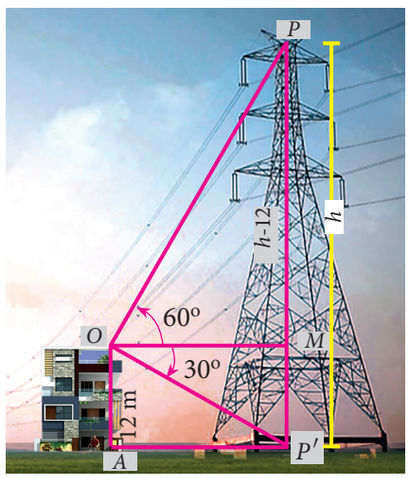


Fig. 6.27

In the right angled ΔOMP, $\dfrac{MP}{OM} = \tan 60°$

$$\Rightarrow \quad \frac{h-12}{OM} = \sqrt{3}$$

$$OM = \frac{h-12}{\sqrt{3}} \qquad \ldots(1)$$

In the right angled $\Delta OMP'$, $\dfrac{MP'}{OM} = \tan 30°$

$$\Rightarrow \quad \frac{12}{OM} = \frac{1}{\sqrt{3}}$$

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